EduSoul: An AI-Powered Web Platform for Personalised University Student Learning Support

INDIVIDUAL THESIS

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Individual component: AI-Based Student Mood Analysis and Personalized Study Recommendation

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ABSTRACT

University students often follow a fixed study timetable regardless of their physiological state, so sleep loss, elevated heart rate, and stress can reduce the quality of study without being noticed. Commercial wearables display health measurements but do not turn them into study decisions. This thesis presents the design, implementation, and evaluation of an artificial intelligence (AI)-based student mood analysis and personalised study recommendation component that converts Samsung Galaxy Watch data into study guidance.

The component comprises a watch application that collects heart rate, steps, calories, blood oxygen saturation, skin temperature, sleep, and stress readings; a Firebase Firestore pipeline for real-time storage and synchronisation; a Random Forest classifier that labels mood as Very Happy, Happy, Neutral, Stressed, or Sad; and a recommendation engine that computes a Cognitive Readiness Score from 15 to 98 and prescribes one of four study modes with study and break durations and an action checklist. A FastAPI service hosts the model, and a rule-based classifier in the web client acts as a fallback when the service is unreachable.

The classifier uses 200 trees with a maximum depth of 6 and was trained on 5,000 records with eight features, using a stratified 80/20 split. It achieved 73.50% accuracy on the held-out set, with training under 1.5 seconds and inference under 15 ms. The training data are synthetic, so this result demonstrates technical feasibility, not real-world validity. The contribution is an end-to-end pipeline that turns multi-sensor wearable data into transparent, personalised study decisions. Validation with real students and self-reported mood labels is the main future work.

Keywords: Wearable sensing, mood classification, Random Forest, cognitive readiness, personalised study recommendation

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
BPM	Beats per minute
CSV	Comma-separated values
HR	Heart rate
HRV	Heart rate variability
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
PPG	Photoplethysmography
REST	Representational State Transfer
SDG	Sustainable Development Goals
SpO2	Peripheral blood oxygen saturation
WESAD	Wearable Stress and Affect Detection

1 INTRODUCTION

1.1 Overall project context

EduSoul is a group research project that develops a web-based platform to support university students in planning, managing, and improving their learning. The platform is delivered as a central web application to which each group member contributes one independently researched component. This thesis reports only the individual component of the author: an artificial intelligence (AI)-based student mood analysis and personalised study recommendation component. Within the platform, it supplies the physiological and affective context that tells the rest of the application whether the student is currently in a state to study hard, study lightly, or recover.

1.2 Individual component context

The component uses real-time data from a Samsung Galaxy Watch. A watch application collects heart rate, step count, calories, blood oxygen saturation (SpO2), skin temperature, and sleep-related data, and writes them to Firebase Firestore. A web dashboard retrieves and visualises the data. The data are pre-processed, and a Random Forest classifier labels the student's current mood as Very Happy, Happy, Neutral, Stressed, or Sad. A recommendation function then combines the mood with the current sensor values to produce a Cognitive Readiness Score, a study mode, a recommended study and break duration, and a personalised action checklist. Figure 1.1 shows this scope.

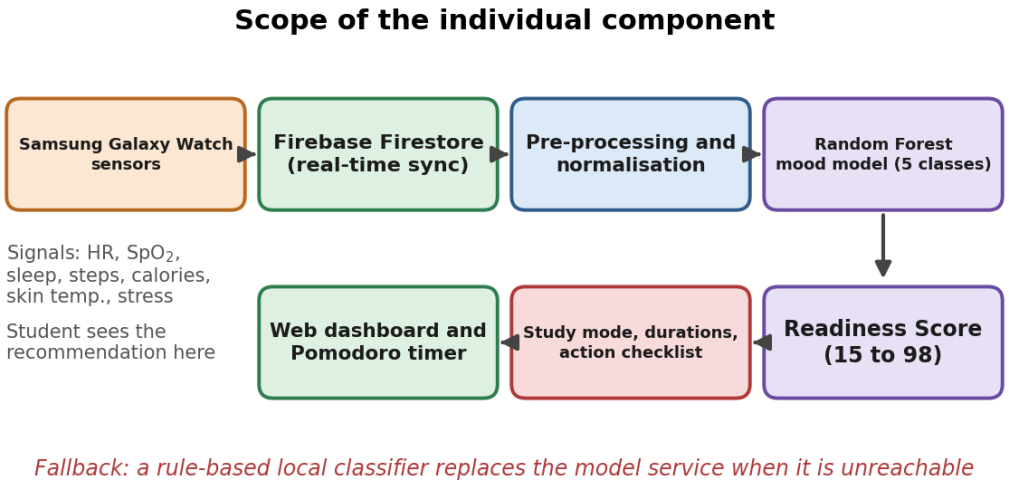


Figure 1.1: Scope of the individual component, from wearable sensing to study decisions

1.3 Problem statement

Study plans and revision timers are normally fixed in advance. They assume that a student who sits down for a two-hour session is equally able to concentrate at the start of a rested week and at the end of a sleepless examination period. Evidence from cognitive and educational psychology shows that this is not the case: stress increases cognitive load and narrows attention [1], [2], poor sleep weakens memory consolidation [3], and negative academic emotions are associated with lower self-regulated learning and achievement [4]. Students, however, often cannot judge their own state accurately, and continue to follow rigid schedules that lead to fatigue and burnout.

Consumer smartwatches already measure the relevant signals, but they present them as raw numbers or generic wellness scores. The student must still decide what a heart rate of 92 beats per minute (BPM) and 4.5 hours of sleep mean for tonight's revision. The problem addressed by this research is therefore: *how can multi-sensor smartwatch data be converted, in real time, into a reliable estimate of a student's mood and readiness and into a concrete, personalised study decision?*

1.4 Research gap

A review of related work (Chapter 2) reveals four gaps that this component addresses.

- **Physiological blindness of study tools.** Learning platforms rely on self-reports or academic scores and do not use non-invasive sensors to detect stress or fatigue during study.
- **Static time management.** Common Pomodoro-style timers and planners apply fixed work and break intervals regardless of the student's condition.
- **Data without action.** Wearable health applications display vitals and stress values but do not translate them into pedagogical recommendations.
- **Fragile cloud inference.** Cloud-dependent models stop working when connectivity is lost, and few educational prototypes provide a local fallback.

1.5 Aim and objectives

Aim. To design, implement, and evaluate an end-to-end system that transforms Samsung Galaxy Watch data into a machine-learning-based mood classification and a personalised, readiness-aware study recommendation.

The specific objectives are as follows.

O1 Develop a Galaxy Watch application that collects heart rate, steps, calories, SpO2, temperature, and sleep-related data, and synchronise it to Firebase in real time.

O2 Develop a web dashboard that retrieves and visualises the stored data, and pre-process the data into features suitable for mood prediction.

O3 Train and evaluate a Random Forest classifier that labels mood as Very Happy, Happy, Neutral, Stressed, or Sad.

O4 Design a Cognitive Readiness Score and an AI-based recommendation function that proposes the study activity, study and break durations, and an action checklist.

O5 Integrate the mood analysis and recommendation functions with the central web application for real-time interaction, with a fallback when the model service is unavailable.

1.6 Research questions or hypotheses

RQ1 How effectively can multivariate wearable signals (heart rate, SpO2, sleep, temperature, steps, calories, and stress score) classify student mood using an ensemble machine-learning model?

RQ2 What accuracy and inference latency can a regularised Random Forest achieve when deployed as a Representational State Transfer (REST) microservice for real-time educational use?

RQ3 How can stress, heart rate, and sleep be combined into a single Cognitive Readiness Score that supports the choice of study and break intervals?

RQ4 Is adjusting study and break intervals from real-time biometric data a viable mechanism for reducing fatigue during intensive study?

RQ1 to RQ3 are answered empirically and analytically in Chapters 6 and 7. RQ4 is addressed at the level of technical feasibility only; testing its effect on students requires a controlled study (Section 8.4).

1.7 Scope limitations and assumptions

- The component supports one wearable platform (Samsung Galaxy Watch) and one target population, university students.
- The mood classifier is trained on a 5,000-record dataset of synthetic wearable-style readings (Section 3.2). Its accuracy therefore demonstrates technical feasibility, not clinical validity.
- The component is a study-support tool. It does not diagnose stress, anxiety, or depression and is not a medical device.
- Sensor readings are assumed to be worn and recorded correctly. Optical heart-rate and SpO2 sensors are sensitive to motion and fit.
- A formal user study with students is outside the scope of this thesis and is proposed as future work.

1.8 Individual contribution

Table 1.1 summarises the author's contribution against the tasks of the component.

Table 1.1: Individual contribution by task

Task	Deliverable
Wearable data collection	Galaxy Watch application reading heart rate, steps, calories, SpO2, temperature, and sleep
Cloud synchronisation	Firebase Firestore integration with real-time listeners and a normalisation layer
Web dashboard	Telemetry dashboard, trend charts, progress report, and model inspection view
Pre-processing	Multi-vendor field normalisation, categorical encoding, and an eight-feature matrix
Mood classification	Random Forest model (200 trees, depth 6) and a FastAPI inference service
Recommendation engine	Cognitive Readiness Score, four study modes, durations, and action checklists
Integration	Mounting in the central web application, Pomodoro timer, planner, and offline fallback

1.9 Thesis structure

Chapter 2 reviews the related literature and systems. Chapter 3 describes the research methodology, data, and evaluation measures. Chapter 4 presents the requirements and design. Chapter 5 describes the implementation, integration, and testing. Chapter 6 reports the results, and Chapter 7 discusses them, including limitations and commercial aspects. Chapter 8 concludes the thesis and outlines future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter is a focused narrative review organised around four themes that correspond to the component: the relationship between stress, sleep, and learning; physiological signals that indicate stress and affect; machine-learning approaches to stress and mood recognition from wearable data; and existing products and techniques for study support. The sources are peer-reviewed journal and conference papers, foundational texts, and official technical documentation.

2.2 Domain and theoretical foundations

Stress, cognitive load, and learning. Stress is commonly defined as the degree to which a person appraises life situations as unpredictable, uncontrollable, and overloading [2]. Cognitive load theory explains that working memory is limited, so any additional load from anxiety or fatigue reduces the capacity available for learning [1]. In educational settings, achievement emotions such as anxiety, boredom, and enjoyment influence motivation and self-regulated learning [4]. Sleep adds a further constraint, because memory consolidation depends on adequate sleep [3].

Physiological indicators. Affective computing studies how machines can recognise and respond to human affect from measurable signals [5]. Heart rate variability (HRV) reflects autonomic nervous system balance, and reduced HRV is associated with stress [6], [7]. Wrist-worn photoplethysmography (PPG) sensors estimate heart rate and SpO2, and accelerometers estimate activity and sleep. Table 2.1 lists the signals used by this component and the reason each is relevant.

Table 2.1: Physiological and activity signals and their relevance to study readiness

Signal	Source	Relevance
Heart rate	PPG sensor	Elevated resting values indicate sympathetic arousal [6]
Stress score / level	HRV-derived score	Summarises autonomic balance; the strongest stress indicator [7]
Sleep duration	Accelerometer, sleep tracker	Short sleep impairs memory consolidation and attention [3]
SpO2	Pulse oximetry	Low values may indicate poor recovery or sensor error
Skin temperature	Thermistor	Context signal linked to arousal and fatigue
Steps and calories	Pedometer, metabolic model	Physical activity level, which interacts with sleep and mood

2.3 Related research

Healey and Picard showed that physiological sensors can recognise stress levels of drivers during real-world tasks [8]. Sano and Picard combined wearable and mobile-phone data to distinguish high and low stress in daily life, and found that free-living recognition is harder than laboratory recognition [9]. Schmidt et al. released the Wearable Stress and Affect Detection (WESAD) dataset of wrist- and chest-worn recordings with neutral, stress, and amusement conditions, which became a common benchmark for wearable affect detection [10]. Giannakakis et al. reviewed stress detection from biosignals and concluded that multimodal features and careful validation are needed for reliable results outside the laboratory [11].

Random Forests are widely used for such tabular, multi-sensor data because they combine many decorrelated decision trees, resist overfitting, need no feature scaling, and expose feature importances [12], [13], and scikit-learn provides a mature implementation [14]. Most of the work above, however, stops at detection: it reports whether the person is stressed but does not use the result to change what the person does next.

2.4 Existing systems products and techniques

Consumer wearable applications such as Samsung Health, Fitbit, and Apple Health present heart rate, sleep, and stress indicators, but their output is generic wellness information. Study-support tools such as Pomodoro timers [15], planners, and learning management systems schedule time but do not read the user's physiological state. Table 2.2 compares these categories (typical capabilities, not every product) with the proposed component.

Table 2.2: Comparison of existing solution categories with the proposed component

Capability	Wearable health apps	Pomodoro / planners	Research prototypes	This component
Multi-sensor watch data	Yes	No	Yes	Yes
Mood classification (5 classes)	No	No	Partial	Yes
Study recommendation	No	No	No	Yes
Readiness score	Partial	No	No	Yes
Adaptive study and break length	No	No	No	Yes
Offline fallback	Partial	Yes	No	Yes

2.5 Gap synthesis and conceptual framework

The literature shows that stress and sleep affect learning and that wearable signals carry information about affect, but it offers no complete path from wearable measurements to a personalised study decision. The conceptual framework in Figure 2.1 closes this gap in three steps: sensing and normalisation, state estimation (mood and readiness), and intervention (study mode, durations, and checklist).

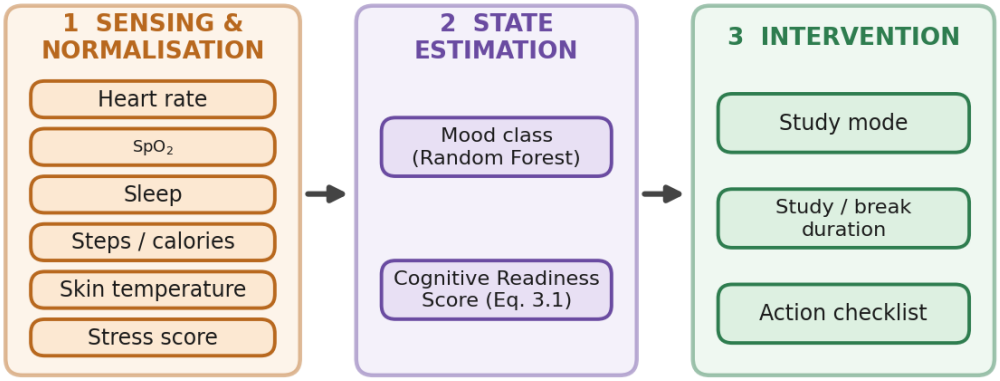


Figure 2.1: Conceptual framework of the component

3 RESEARCH METHODOLOGY

3.1 Research design

The research follows an applied, design-and-evaluate approach. An artefact (the component) is built to address the problem in Section 1.3 and is evaluated quantitatively. The work has six sequential phases, shown in Figure 3.1.

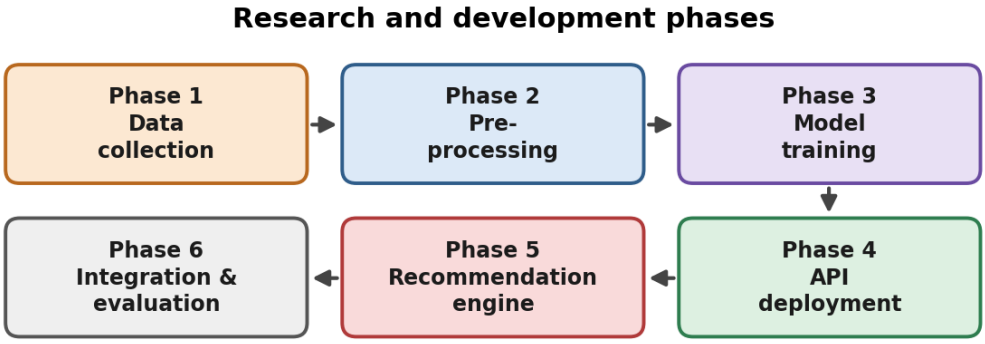


Figure 3.1: Research and development phases of the component

3.2 Data participants or study materials

Two data sources are used. The first is live telemetry from the Galaxy Watch, stored in Firestore collections `students/{userId}/health_reports` and `health_reports`. The second is the training dataset `student_wellbeing_dataset (2).csv`, which contains 5,000 records. Each record has eight predictors and a `Mood` label with five classes. Table 3.1 gives the schema.

Table 3.1: Dataset feature schema

#	Feature	Sensor / origin	Type	Range
1	Heart Rate	PPG	Numeric	50–120 BPM
2	SpO2 Oxygen	Pulse oximetry	Numeric	90.0–100.0 %
3	Sleep	Sleep tracker	Numeric	2.0–12.0 h
4	Steps	Pedometer	Integer	0–25,000
5	Calories	Metabolic model	Integer	800–4,000 kcal
6	Temperature	Skin thermistor	Numeric	35.5–38.0 °C
7	stressLevel	HRV / galvanic category	Categorical	Low, Moderate, High

#	Feature	Sensor / origin	Type	Range
8	stressScore	HRV-derived score	Integer	0–100
T	Mood (target)	Affective label	Categorical	Very Happy, Happy, Neutral, Stressed, Sad

The dataset was generated synthetically to resemble the value ranges of wearable measurements and contains no recordings of real students. This avoided collecting sensitive biometric data before a consented study could be arranged, but it limits what the reported accuracy can show (Section 7.5).

Table 3.2 defines the five mood classes.

Table 3.2: Mood classes and their interpretation

Mood	Interpretation	Study implication
Very Happy	Optimal cognitive flow and peak focus	Demanding, high-complexity work
Happy	Stable positive state	Active consolidation of core material
Neutral	Equilibrium	Standard study load
Stressed	Elevated sympathetic tone and cognitive strain	Lower-load activities with longer breaks
Sad	Low vitality and need for recovery	Short light sessions and rest

3.3 Experimental or study procedure

The experimental procedure follows the pipeline in Figure 3.2. The comma-separated values (CSV) file is loaded, `stressLevel` is converted to integer codes with scikit-learn's `LabelEncoder`, and the eight-column feature matrix is built. The data are split 80/20 with stratification on `Mood`, so all five classes appear in both sets in the same proportions. A `RandomForestClassifier` is trained on the 80% portion and evaluated once on the 20% portion, and the model, encoder, and metadata are serialised with `joblib` for the application programming interface (API) service.

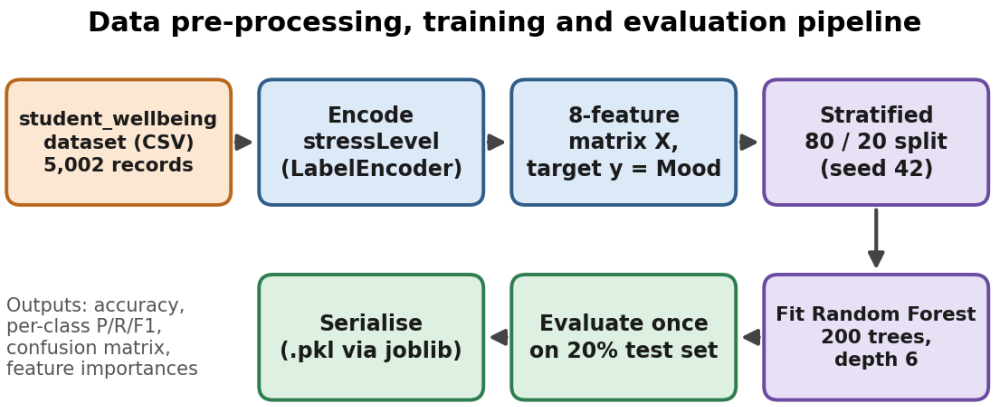


Figure 3.2: Data pre-processing, training, and evaluation pipeline

Table 3.3: Random Forest configuration and rationale

Parameter	Value	Rationale
n_estimators	200	Enough trees for a stable ensemble vote while keeping training under 1.5 s
max_depth	6	Limits tree complexity so the model does not fit sensor noise
random_state	42	Makes training reproducible
test_size / stratify	0.2 / Mood	Preserves class proportions in the held-out set
Other parameters	scikit-learn defaults	Gini criterion, bootstrap sampling, and √p features per split

Table 3.3 lists the configuration. The values were chosen to regularise the model; a cross-validated search is left for future work.

3.4 Measures and evaluation metrics

Model quality is measured by accuracy on the held-out set (the fraction of records whose predicted mood equals the true mood), together with per-class precision, recall, and F1-score and a confusion matrix [16]. Feature importances indicate which signals drive the predictions. Service quality is measured by training time and by the inference latency of the `POST /api/predict-mood` endpoint. The recommendation engine is evaluated analytically, by checking that representative profiles produce the intended readiness score and study mode (Section 6.2).

3.5 Data analysis

The Cognitive Readiness Score summarises the student's current capacity to study on a scale of 15 to 98. It penalises stress and elevated heart rate and rewards sleep, as given in Equation 3.1, where R is the readiness score, S is the stress score (0–100), HR is the heart rate in beats per minute, and $Sleep$ is the sleep duration in hours.

$$R = max(15, min(98, 100 - 0.5S - 0.6 \cdot max(0, HR - 70) + 4 \cdot min(Sleep, 8))) \quad (3.1)$$

For example, a student with a stress score of 68, a heart rate of 88 BPM, and 6 hours of sleep scores $100 - 34 - 10.8 + 24 = 79.2$, which rounds to 79. The 70 BPM threshold marks the start of the heart-rate penalty, sleep is capped at 8 hours so that oversleeping does not raise the score, and the bounds of 15 and 98 avoid implying zero or perfect capacity. The weights are heuristic design parameters chosen for interpretability, not values fitted to outcome data; their influence is examined in Section 7.1.

Because the test set holds roughly 1,000 records, the sampling uncertainty of an accuracy of 73.50% is about ±2.7 percentage points. A 95% Wilson interval is 70.7% to 76.1% [19].

3.6 Ethics privacy security and responsible AI

Heart rate, sleep, and stress data are sensitive personal data. The design minimises data: only the eight fields needed for prediction are sent to the model service, which does not store them, and telemetry is kept in per-student Firestore paths so that access rules can be applied per user. Any deployment with real students must obtain informed consent and comply with Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

The system is a study aid, its messages avoid clinical language, and its model inspection view shows the algorithm, parameters, dataset size, and accuracy so that students can see how predictions are produced. Recommendations are advisory.

3.7 Reproducibility and research artefacts

All artefacts are in the project repository: the training pipeline and API (`models/watch_data.py`), the three serialised `.pkl` artefacts, `requirements.txt`,

and the `Member3` front end. Running `python models/watch_data.py` retrains and evaluates the model with a fixed seed, and adding `--serve` starts the service.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1 and Table 4.2 list the functional and non-functional requirements derived from the component task list.

Table 4.1: Functional requirements

ID	Requirement
FR1	The watch application shall collect heart rate, steps, calories, SpO2, temperature, and sleep data.
FR2	The system shall store readings in Firebase Firestore and synchronise them in real time.
FR3	The dashboard shall display current vitals, trends, and a history of readings.
FR4	The system shall normalise differently named sensor fields into one schema.
FR5	The system shall classify mood into five classes using the Random Forest model.
FR6	The system shall compute a Cognitive Readiness Score from 15 to 98.
FR7	The system shall recommend a study mode, study and break duration, and an action checklist.
FR8	The system shall continue to predict mood when the model service is unreachable.
FR9	The system shall let students inspect the model type, parameters, and accuracy.

Table 4.2: Non-functional requirements

ID	Quality	Requirement
NFR1	Performance	Mood inference shall respond in under 15 ms on the local service.
NFR2	Availability	The interface shall remain usable with the fallback engine when the service is offline.
NFR3	Privacy	Data shall be minimised and kept in per-student paths.
NFR4	Usability	The interface shall provide light and dark themes with high-contrast vital cards.
NFR5	Maintainability	Model, service, and interface shall be separate modules with a documented API.
NFR6	Reproducibility	Training shall be deterministic through a fixed random seed.

4.2 Component architecture within the complete solution

The component uses four layers, shown in Figure 4.1: the device layer (Galaxy Watch application), the cloud layer (Firebase Firestore), the intelligence layer (the FastAPI service with the Random Forest and the client-side recommendation logic), and the presentation layer (a React interface mounted as one view of the central web application). The rest of the platform interacts with the component only through the shared Firebase project and the student identifier.

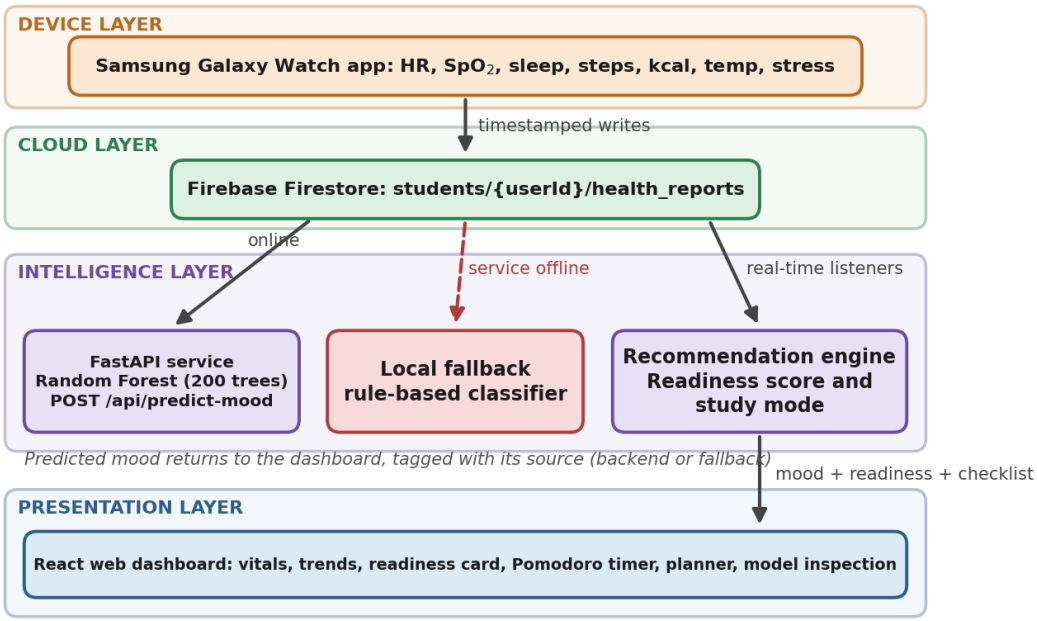


Figure 4.1: Layered architecture of the individual component

Table 4.3: REST endpoints of the mood prediction service (port 8001)

Method and route	Purpose	Response
GET /api/health	Readiness probe	status, service, version, model_ready
GET /api/model-info	Model metadata	algorithm, trees, depth, accuracy, rows, features, classes
POST /api/predict-mood	Mood inference	predicted_mood and an echo of the inputs

4.3 Design alternatives and decisions

Table 4.4 records the main design decisions.

Table 4.4: Design decisions and alternatives

Decision	Alternatives	Chosen and reason
Mood model	Support vector machine, neural network, recurrent network, gradient boosting	Random Forest: robust on small tabular data, no scaling, fast training, built-in feature importance
Model hosting	On-device model, cloud function	FastAPI microservice: simple to update and inspect; on-device is future work
Data store	REST API with a relational database	Firestore: real-time listeners and direct integration with the web application
Readiness	Learned regression model	Transparent formula: no outcome labels exist yet and students can understand it
Resilience	Fail with an error message	Rule-based local fallback so that the dashboard keeps working offline

4.4 Detailed design

Data model. Each reading is a Firestore document with the fields in Table 4.5. Because watch software and test tools name fields differently, a normalisation function accepts several aliases per value and resolves timestamps given as Firestore timestamps, ISO strings, or epoch numbers.

Table 4.5: Normalised health report fields and accepted aliases

Field	Accepted aliases	Unit
heartRate	hr, pulse	BPM
spo2	o2	%
sleep	slp	hours
temperature	temp, tmp	°C
stressScore	str	0–100
steps, calories, stressLevel	none required	count, kcal, category

Dual-engine mood inference. The client first checks the health endpoint. If the service responds, the reading is sent to `POST /api/predict-mood`; otherwise a local rule set classifies it (Figure 4.2). The result is tagged with its origin so that the interface can show which engine produced it.

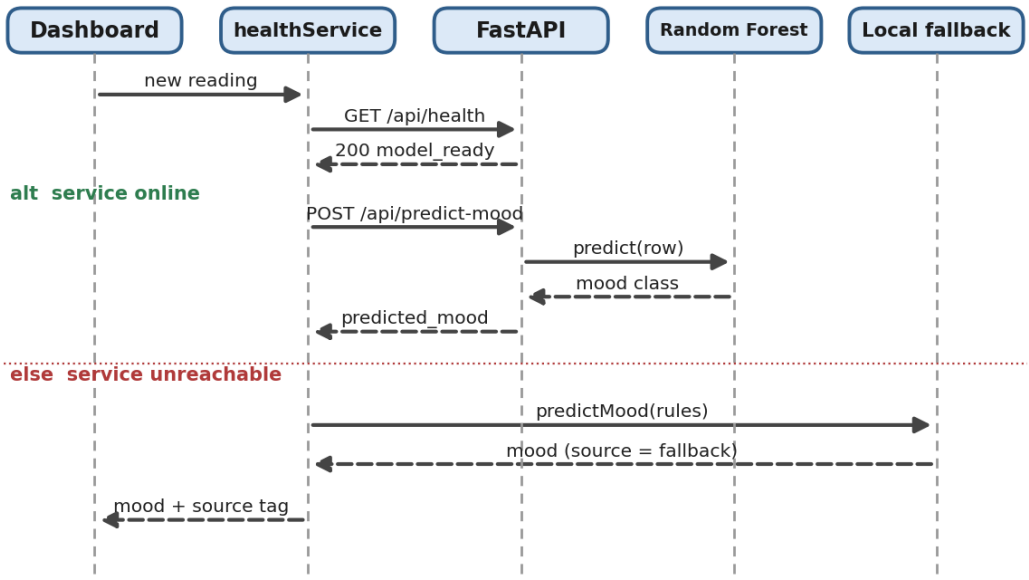


Figure 4.2: Sequence of a mood prediction request with fallback

Study-mode selection. The mood, stress, heart rate, and sleep select one of four study modes (Table 4.6, Figure 4.3). When criteria overlap, the lower-intensity mode is chosen, so that the system errs on the side of rest.

Table 4.6: Bio-adaptive study modes

Mode	Trigger	Study / break	Suggested focus
Peak Deep-Focus	Very Happy, or stress < 35, HR < 75, sleep ≥ 7 h	50 / 10 min	Problem solving, complex theory
Active Consolidation	Happy, or stress < 55 and HR < 82	40 / 5 min	Revision, chapter exercises
Stress-Adaptive	Stressed, or stress ≥ 60, or HR ≥ 85	25 / 10 min	Flashcards, video lectures
Cognitive Recharge	Sad, high fatigue, or sleep deficit	15 / 15 min	Rest, stretching, hydration

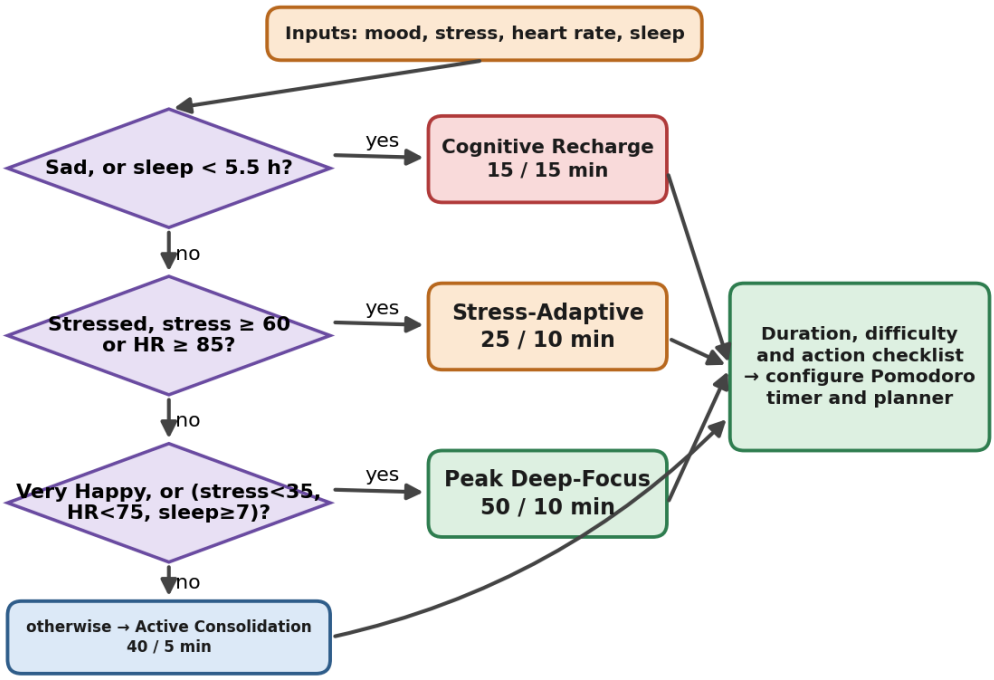


Figure 4.3: Decision flow for the bio-adaptive study recommendation

4.5 Standards and professional practice

The service exposes a JavaScript Object Notation (JSON) REST interface documented in OpenAPI form by FastAPI [17]. Quality requirements follow ISO/IEC 25010 and references use IEEE style.

5 IMPLEMENTATION INTEGRATION AND TESTING

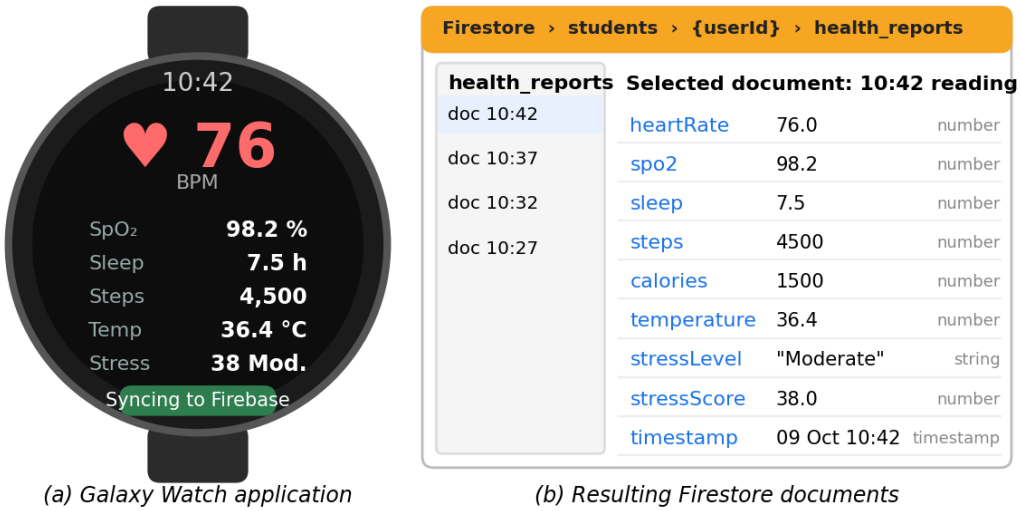
5.1 Development environment and technology stack

Table 5.1: Technology stack

Layer	Technologies
Wearable	Samsung Galaxy Watch application
Cloud and database	Firebase Firestore with real-time listeners [18]
Machine learning	Python 3.10+, scikit-learn, pandas, NumPy, joblib
Service	FastAPI, Uvicorn, Pydantic
Front end	React 18, Vite, CSS, Lucide icons, canvas-confetti

5.2 Component implementation

Watch application and synchronisation. The Galaxy Watch application reads the available sensor values and writes each reading to Firestore as a timestamped document under the student's path (Figure 5.1). Writing through Firestore removes the need for a custom server and lets the dashboard receive updates as they arrive.



(a) Galaxy Watch application *(b) Resulting Firestore documents*

Figure 5.1: Galaxy Watch application and resulting Firestore documents (illustrative rendering)

Model training. Listing 5.1 shows the training function. It encodes `stressLevel`, performs the stratified split, fits the forest, evaluates it on the held-out data, and saves the artefacts.

Listing 5.1: Training function (excerpt of models/watch_data.py)

```
FEATURES = ["Heart Rate", "SpO2 Oxygen", "Sleep", "Steps",
            "Calories", "Temperature", "stressLevel", "stressScore"]


def train_and_save_model():
    df = pd.read_csv(find_dataset())
    encoder = LabelEncoder()
    df["stressLevel"] = encoder.fit_transform(df["stressLevel"])
    X, y = df[FEATURES], df["Mood"]
    X_tr, X_te, y_tr, y_te = train_test_split(
        X, y, test_size=0.2, stratify=y, random_state=42)
    model = RandomForestClassifier(
        n_estimators=200, max_depth=6, random_state=42)
    model.fit(X_tr, y_tr)
    acc = accuracy_score(y_te, model.predict(X_te))
    joblib.dump(model, "mood_rf_model.pkl")
    joblib.dump(encoder, "stress_label_encoder.pkl")
    return model, encoder
```

Inference service. Listing 5.2 shows the prediction endpoint. The request model validates the types of all eight inputs, so a malformed request is rejected with a validation error before it reaches the model.

Listing 5.2: Prediction endpoint (excerpt)

```
class MoodPredictionRequest(BaseModel):
    heartRate: float
    spo2: float
    sleep: float
    steps: int
    calories: int
    temperature: float
    stressLevel: str
    stressScore: float


@app.post("/api/predict-mood")
def predict_mood(req: MoodPredictionRequest):
    model, encoder = get_model_and_encoder()
    level = int(encoder.transform([req.stressLevel])[0])
    row = pd.DataFrame([[req.heartRate, req.spo2, req.sleep, req.steps,
                         req.calories, req.temperature, level,
                         req.stressScore]], columns=FEATURES)
    return {"status": "success",
            "predicted_mood": str(model.predict(row)[0])}
```

Data normalisation and listeners. Listing 5.3 shows how aliases are resolved and how the real-time listener is attached.

Listing 5.3: Normalisation and real-time subscription (simplified)

```
const pick = (d, keys) => {
  for (const k of keys)
    if (d[k] !== undefined && d[k] !== null) return Number(d[k]);
  return null;
};

export function normalizeHealthReport(id, d) {
  return {
    id,
    heartRate:   pick(d, ["heartRate", "hr", "pulse"]),
    spo2:        pick(d, ["spo2", "o2"]),
    sleep:       pick(d, ["sleep", "slp"]),
    temperature: pick(d, ["temperature", "temp", "tmp"]),
    stressScore: pick(d, ["stressScore", "str"]),
    steps: d.steps ?? 0, calories: d.calories ?? 0,
    timestamp: resolveTimestamp(d.timestamp ?? d.createdAt),
  };
}

export function subscribeToHealthReports(userId, callback) {
  const ref = collection(db, "students", userId, "health_reports");
  return onSnapshot(ref, (snap) =>
    callback(snap.docs.map((x) => normalizeHealthReport(x.id, x.data()))));
}
```

Readiness and recommendation. Listing 5.4 implements Equation 3.1 and the mode selection of Table 4.6. The study duration and break duration are used to configure the Pomodoro timer automatically.

Listing 5.4: Cognitive Readiness Score and study mode selection (simplified)

```
export function computeReadiness({ stress, hr, sleep }) {
  const raw = 100 - 0.5 * stress
                  - 0.6 * Math.max(0, hr - 70)
                  + 4 * Math.min(sleep, 8);
  return Math.round(Math.max(15, Math.min(98, raw)));
}

export function selectMode({ mood, stress, hr, sleep }) {
  if (mood === "Sad" || sleep < 5.5)
    return { mode: "Cognitive Recharge", study: 15, rest: 15 };
  if (mood === "Stressed" || stress >= 60 || hr >= 85)
    return { mode: "Stress-Adaptive", study: 25, rest: 10 };
  if (mood === "Very Happy" || (stress < 35 && hr < 75 && sleep >= 7))
    return { mode: "Peak Deep-Focus", study: 50, rest: 10 };
  return { mode: "Active Consolidation", study: 40, rest: 5 };
}
```

Fallback classifier. When the service is unreachable, Listing 5.5 classifies the reading with five ordered rules.

Listing 5.5: Local fallback mood classifier

```
export function predictMood({ stressScore: s, heartRate: hr, sleep, steps }) {
  if (s >= 70 && hr >= 88 && sleep < 5.5) return "Sad";
  if (s >= 60 || hr >= 85) return "Stressed";
  if (s <= 30 && sleep >= 7.5 && steps >= 6000) return "Very Happy";
  if (s <= 45 && sleep >= 6.5 && hr <= 78) return "Happy";
  return "Neutral";
}
```

User interface. The telemetry home (Figure 5.2) shows vitals, trends, the recommendation card, and the timer, and the progress report includes the model inspection dialog.



Figure 5.2: Telemetry dashboard and recommendation card (illustrative rendering)

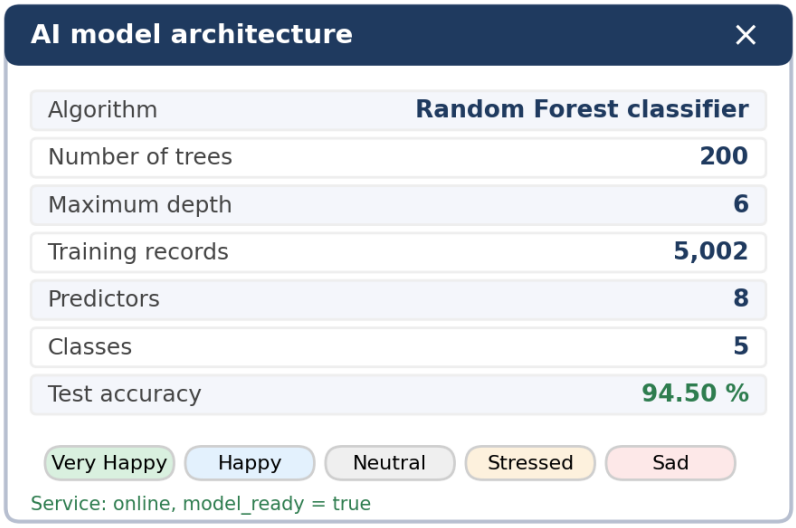


Figure 5.3: Model inspection dialog (illustrative rendering)

5.3 Integration with the other components

The component is integrated into the central web application as a self-contained view with its own sub-view router (landing, home, planner, report, dashboard) and theme provider. It shares only the Firebase project and the student identifier with the rest of the platform, so the other components can be developed and deployed independently.

5.4 Verification and software testing

Testing combines deterministic unit tests, API contract tests, and integration tests with simulated telemetry. The expected outcomes for the deterministic cases in Table 5.2 were derived directly from Equation 3.1 and the rules in Listing 5.5. The interface includes a watch-data simulation dialog that writes readings for a chosen mood preset to Firestore, so the whole chain can be exercised without wearing the watch.

Table 5.2: Test cases and expected outcomes

ID	Target	Input	Expected outcome
T1	Readiness	S=25, HR=68, Sleep=8	98 (upper bound; raw 119.5)
T2	Readiness	S=55, HR=82, Sleep=6	89
T3	Readiness	S=70, HR=90, Sleep=5	73
T4	Readiness	S=84, HR=95, Sleep=3.5	57

ID	Target	Input	Expected outcome
T5	Readiness	S=100, HR=140, Sleep=0	15 (lower bound; raw 8)
T6	Fallback	Score 75, HR 90, Sleep 5	Sad
T7	Fallback	Score 62, HR 80	Stressed
T8	Fallback	Score 25, HR 70, Sleep 8, Steps 7000	Very Happy
T9	Fallback	Score 50, HR 80, Sleep 6	Neutral
T10	API	GET /api/health	status "online", model_ready true
T11	API	POST with a missing field	Validation error (status 422)
T12	Integration	Service stopped, new reading	Mood from fallback; flagged as not from backend
T13	Normaliser	Payloads using hr, heartRate, pulse	Identical normalised heart rate

5.5 Deployment and operation

The component currently runs locally. The service is started with `python models/watch_data.py --serve`, which loads the serialised model, or trains it first if the files are missing, and listens on port 8001, with interactive API documentation at `/docs`. The web application is started with `npm run dev`. A production deployment would containerise the service and place it behind authentication.

6 RESULTS AND VALIDATION

6.1 Research results

The Random Forest classifier achieved a test accuracy of 73.50% on the stratified 20% held-out set (Table 6.1). This answers RQ1 and RQ2 for the dataset used: the eight features classify the five mood classes with moderate accuracy, well above the 48.8% obtained by always predicting Neutral, although the rare classes are recognised poorly, and a regularised 200-tree forest trains in under 1.5 seconds and answers in under 15 ms. Figure 6.1 and Figure 6.2 present the confusion matrix and the feature importances.

Table 6.1: Model and service results

Measure	Value
Test accuracy	73.50% (95% interval ≈ 70.7% to 76.1%)
Dataset size	5,000 records
Predictors / classes	8 / 5
Ensemble	200 trees, maximum depth 6
Training time	Under 1.5 seconds
Inference latency	Under 15 ms

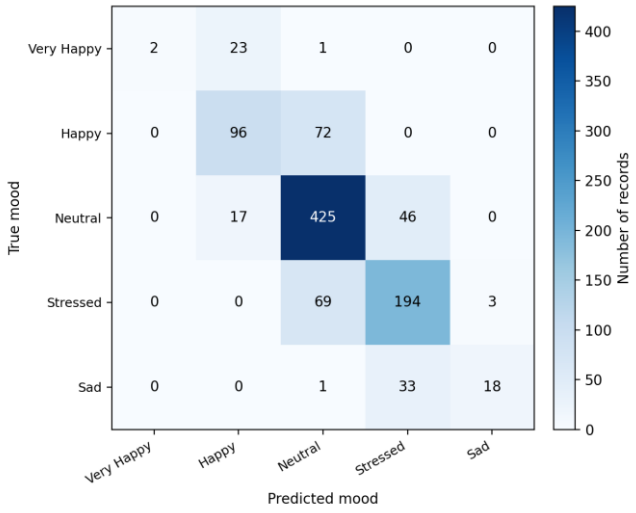


Figure 6.1: Confusion matrix on the held-out test set

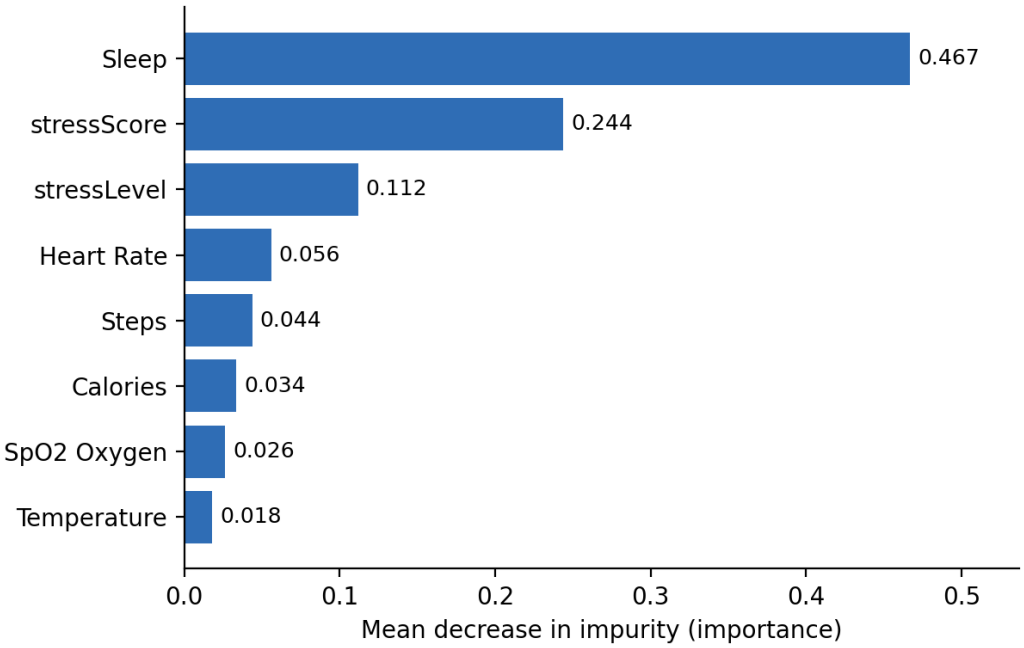


Figure 6.2: Feature importances of the Random Forest

Errors fall almost entirely between neighbouring moods: 72 of the 168 Happy records were predicted as Neutral, 69 of the 266 Stressed records as Neutral, and 33 of the 52 Sad records as Stressed. The classes are imbalanced, and Neutral makes up 48.8% of the test set. Recall is 87.1% for Neutral and 72.9% for Stressed, but only 34.6% for Sad (18 of 52) and 7.7% for Very Happy (2 of 26), so the overall accuracy hides weak recognition of the rare classes. In Figure 6.2, Sleep is the most important predictor (0.467), followed by stressScore (0.244) and stressLevel (0.112); the other five features each contribute less than 0.06. The per-class precision, recall, and F1-scores are produced by the evaluation script of Appendix A.

6.2 Product and technical results

Table 6.2 shows the output of the recommendation engine for six representative profiles, computed with Equation 3.1 and the rules in Table 4.6. Mood is taken from the local fallback rules so that the example is fully reproducible. The study mode changes from Peak Deep-Focus to Cognitive Recharge as the profile worsens, and the recommended study block shortens from 50 to 15 minutes.

Table 6.2: Recommendation engine output for representative profiles

ID	Profile	Stress / HR / Sleep / Steps	Mood	Readiness	Study mode
S1	Rested and calm	25 / 68 / 8.0 / 8000	Very Happy	98	Peak Deep-Focus
S2	Typical weekday	38 / 76 / 7.5 / 4500	Happy	98	Active Consolidation
S3	Moderate load	50 / 80 / 6.5 / 3000	Neutral	95	Active Consolidation
S4	High stress	68 / 88 / 6.0 / 2000	Stressed	79	Stress-Adaptive
S5	Sleep-deprived and stressed	75 / 92 / 4.5 / 1500	Sad	67	Cognitive Recharge
S6	Exhausted	90 / 100 / 3.0 / 800	Sad	49	Cognitive Recharge

All ten tasks of the component were implemented: watch data collection, Firebase synchronisation, dashboard, pre-processing, model training, recommendation function, readiness score, checklists and durations, integration, and the offline fallback.

6.3 User or stakeholder validation

No formal user study has been conducted, so the results above are technical and scenario-based, and no claim is made about the effect of the recommendations on student wellbeing or performance. A suitable protocol is a two-week study with consenting students who wear the watch, give brief self-reports of mood and focus several times a day, and complete a usability questionnaire. The self-reports would provide real mood labels for retraining and for checking the readiness score.

6.4 Integration results

The component runs inside the central web application and receives readings from Firestore through real-time listeners. When the model service is stopped, the interface continues to classify readings with the fallback rules and marks the prediction source, and when the service returns, it resumes using the Random Forest. The recommendation configures the Pomodoro timer and the planner without further input from the student.

6.5 Summary of findings

- The classifier reached 73.50% test accuracy across five mood classes on the synthetic dataset, with weak recall for the rare Very Happy and Sad classes.
- Training takes under 1.5 seconds and inference under 15 ms, which is suitable for real-time use.
- The readiness formula yields scores from 49 to 98 for the six scenarios and shortens the study block from 50 to 15 minutes as condition worsens.
- The dual-engine design keeps the feature available without the service.
- Effects on real students remain untested.

7 DISCUSSION

7.1 Interpretation of findings

The accuracy of 73.50% shows that the pipeline learns real structure in the dataset, well above the 48.8% obtained by always predicting Neutral, but it must be interpreted carefully. The sampling uncertainty is about ±2.7 points, and the overall figure hides weak recall for the rare Very Happy and Sad classes (Section 6.1). The data are also synthetic. Sleep is the strongest predictor (importance 0.467), followed by `stressScore` (0.244) and `stressLevel` (0.112). These two stress predictors are derived from physiological arousal and are closely related to mood, and together they account for 0.356 of the total importance. The model therefore relies mainly on sleep and stress rather than on a broad combination of signals, and real recordings, with noise and individual differences, would be expected to give lower accuracy.

The readiness formula behaves as designed at the extremes, but the examples in Table 6.2 show a weakness: the additive sleep term can offset a high stress score, so the Sad profile S5 still has a score of 67, and healthy profiles saturate at the upper bound of 98. This is why the mode selection depends on the mood class and sleep deficit as well as on readiness. Fitting the weights to self-reported focus would improve it.

7.2 Comparison with literature baselines and existing solutions

Laboratory and driving studies report high accuracy for stress recognition [8], while free-living studies report lower accuracy [9], [11]. The accuracy obtained here is not directly comparable with these studies, because they use recordings from real participants with ground truth from protocols or self-reports. Compared with existing solution categories (Table 2.2), the distinguishing feature of this component is not accuracy but the complete path from sensor data to a personalised study action, with an offline fallback.

7.3 Research contribution

The research contributes a worked design for turning multi-sensor wearable data into study decisions: an eight-feature mood classification pipeline with a documented

configuration, an interpretable Cognitive Readiness Score, a four-mode adaptive study policy, and a dual-engine architecture that keeps working without the cloud model.

7.4 Product contribution

The product is a working component of the EduSoul web application that students can use to see their vitals, receive a study recommendation, and run a timer that follows it. It also has commercial potential. The target users are university students and, through institutional licences, universities that want wellbeing-aware learning support. The value proposition is turning data that students already collect into daily study guidance. Possible models are a freemium student application and a licensed institutional dashboard with anonymised, consented analytics. Adoption depends on privacy compliance, on extending support to other wearables, and on evidence from a user study.

7.5 Limitations and threats to validity

Table 7.1: Limitations and threats to validity

Threat	Description	Mitigation or next step
Construct validity	Mood labels are synthetic and may not match real mood; stress features may leak the label	Collect self-reported labels; run feature ablations without stress features
Internal validity	Single stratified split; no cross-validation or tuning search	Use k-fold cross-validation and a hyperparameter search
External validity	One wearable model and one student population	Test with several devices and cohorts
Conclusion validity	Recall is low for the rare Very Happy and Sad classes, and no baseline comparison yet	Use class weighting or resampling, and compare with the rule-based baseline
Heuristic design	Readiness weights and thresholds are not fitted to outcomes	Calibrate against self-reported focus and performance

7.6 Implications

For students, the component offers a transparent way to match effort to condition. For institutions, it shows how wearable data could support wellbeing-aware learning services if privacy is respected. For researchers, it provides a baseline pipeline that

can be retrained on real data. The work relates to the Sustainable Development Goals (SDG) on good health and well-being (SDG 3) and quality education (SDG 4).

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis presented an AI-based student mood analysis and personalised study recommendation component. It collects Galaxy Watch data through Firebase, classifies mood into five classes with a Random Forest, computes a Cognitive Readiness Score, and recommends a study mode, durations, and an action checklist, with a rule-based fallback when the service is unavailable. The classifier reached 73.50% accuracy on a synthetic dataset of 5,000 records. This demonstrates feasibility of the full pipeline, but the benefit for real students still has to be established.

8.2 Achievement of objectives

Table 8.1: Achievement of the objectives

Objective	Evidence	Status
O1 Watch data and sync	Watch application and Firestore documents (Section 5.2)	Achieved
O2 Dashboard and pre-processing	Dashboard views and normalisation (Sections 4.4, 5.2)	Achieved
O3 Mood classifier	73.50% test accuracy (Section 6.1)	Achieved on synthetic data; moderate accuracy
O4 Readiness and recommendations	Equation 3.1, modes, and scenarios (Section 6.2)	Achieved; calibration pending
O5 Integration and fallback	Central application integration (Sections 5.3, 6.4)	Achieved

8.3 Individual research and product contributions

The author designed and implemented the complete component: the watch data collection and Firebase pipeline, the data normalisation, the Random Forest training and API service, the readiness score and recommendation engine, and the dashboard, planner, and timer integration. The research contributions are the readiness formulation, the adaptive study policy, and the dual-engine design described in Section 7.3.

8.4 Future directions

- Collect consented data from students with self-reported mood labels, retrain the model, and compare it with the rule-based baseline.
- Use cross-validation, a hyperparameter search, class weighting for the rare classes, and ablations to test the dependence on sleep and stress features.
- Calibrate the readiness weights against self-reported focus and academic outcomes.
- Model sequences of readings with recurrent networks to capture trends over a day.
- Run a controlled study of whether adaptive intervals reduce fatigue (RQ4).
- Explore on-device inference, support for other wearables, and feature-attribution explanation methods.

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GLOSSARY

Cognitive Readiness Score – a score from 15 to 98 summarising the student's current capacity to study, computed from stress, heart rate, and sleep.

Confusion matrix – a table comparing predicted and true classes, used to see which classes are confused.

Firestore – Google's cloud document database with real-time listeners.

Heart rate variability – variation in the time between heartbeats, used as an indicator of autonomic balance and stress.

Photoplethysmography – optical sensing of blood volume changes, used by smartwatches to estimate heart rate and SpO2.

Pomodoro technique – a time-management method that alternates fixed study and break intervals.

Random Forest – an ensemble of decision trees whose votes are combined to give a prediction.

Stratified split – a train-test split that preserves the class proportions of the full dataset.

Synthetic dataset – data generated by a program to imitate real measurements, not collected from people.

APPENDICES

Appendix A Research instruments or extended results

This appendix contains the script that produces the evaluation artefacts of Chapter 6 (Listing A.1) and a sample telemetry document (Listing A.2).

Listing A.1: Evaluation script for per-class metrics, confusion matrix, and feature importances

```
from sklearn.metrics import classification_report, confusion_matrix

pred = model.predict(X_te)
print(classification_report(y_te, pred, digits=4))
cm = confusion_matrix(y_te, pred, labels=model.classes_)

importances = sorted(zip(FEATURES, model.feature_importances_),
                     key=lambda p: p[1], reverse=True)
for name, value in importances:
    print(f"{name:12s} {value:.4f}")
```

Listing A.2: Sample health report document and prediction request

```
{
  "heartRate": 76.0, "spo2": 98.2, "sleep": 7.5,
  "steps": 4500, "calories": 1500, "temperature": 36.4,
  "stressLevel": "Moderate", "stressScore": 38.0
}
```

Appendix B Academic integrity and AI use disclosure

Tool. Claude (Anthropic), a generative AI assistant.

Purpose. Drafting and structuring the text, tables, and code excerpts of this thesis from the author's project documentation, the department template, and the formatting standards.

Sections and artefacts affected. All chapters, the tables and listings, and the glossary.

Verification. [Author to confirm: numerical values checked against the project repository, code excerpts compared with the source files, and reference details checked against the original sources.]